

МОСКОВСКИЙ АВИАЦИОННЫЙ ИНСТИТУТ
(НАЦИОНАЛЬНЫЙ ИССЛЕДОВАТЕЛЬСКИЙ УНИВЕРСИТЕТ)

Институт №8 «Компьютерные науки и прикладная математика»
Кафедра 806 «Вычислительная математика и программирование»

Лабораторная работа №5
по курсу «Операционные системы»

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Условие

Цель работы:

Приобретение практических навыков диагностики работы программного обеспечения.

Задание: При выполнении лабораторных работ по курсу ОС необходимо продемонстрировать ключевые системные вызовы, которые в них используются и то, что их использование соответствует варианту ЛР. По итогам выполнения всех лабораторных работ отчет по данной ЛР должен содержать краткую сводку по исследованию написанных программ.

Вариант: -

Метод решения

Реализованные к каждой лабораторной работе программы запускаются со strace логированием. Программам подаются тестовые данные, ожидается завершение выполнения и программа корректно завершается. Логи сохраняются в указанный при запуске файл формата .txt.

Описание программы

Ключевые строки strace логов к лабораторной работе 1:

```
16293 pipe2([3, 4], 0) = 0
```

```
16293 pipe2([5, 6], 0) = 0
```

```
16293 clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD, child_tidptr=0x7007edb61a10) = 16295
```

Создание двухнаправленного канала для связи родителя с дочерним процессом (pipe1 для нечётных строк) и создание нового независимого процесса. Флаг SIGCHLD отправляет сигнал родителю при завершении дочернего процесса. Каждый процесс получает уникальный PID и независимое адресное пространство.

```
16295 dup2(3, 0) = 0
```

```
16295 dup2(6, 1) = 1
```

Дочерний процесс перенаправляет stdin на конец чтения pipe (теперь читает данные от родителя) и stdout на дескриптор файла 6 (выходной файл "filename.txt").

```
16295 execve("./child ["child "popa"], 0x7ffc371e6c18 /* 49 vars */) = 0
```

Замена образа процесса на программу ./child. Перенаправленные дескрипторы остаются активными в новой программе. Это позволяет child.c читать из pipe и писать в файл без явного открытия этих ресурсов.

Ключевые строки strace логов к лабораторной работе 2:

```
clone3(flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLO
```

Создание потока (не процесса). Флаг CLONE_THREAD указывает, что это поток одного

процесса. CLONE_VM означает общее адресное пространство - все потоки видят одну память. CLONE_SIGHAND означает общие обработчики сигналов.

```
rt_sigaction(SIGRT_1, sa_handler = 0x7dab8ce99530, sa_mask = [], sa_flags = SA_RESTORER|SA_ONSTACK, 0x7dab8ce45330, NULL, 8) = 0
rt_sigprocmask(SIG_UNBLOCK, [RTMINRT_1], NULL, 8) = 0
```

Регистрация обработчика real-time сигнала SIGRT_1. Это необходимо для правильной работы механизма работы потоков в glibc. SIG_UNBLOCK разрешает обработку сигнала в главном потоке.

```
mmap(NULL, 8392704, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0)
mprotect(0x7dab8c600000, 8388608, PROT_READ|PROT_WRITE)
```

Выделение 8 МБ памяти для стека каждого потока. Первый mmap с флагом PROT_NONE создаёт защищённую область, затем mprotect устанавливает права на основную часть стека. Это позволяет ядру обнаружить переполнение стека

```
futex(0x7dab8cdf990, FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 23410, NULL, FUTEX_BITSET_MATCH_ANY) = 0
```

Главный поток ждёт завершения потока с TID=45189. FUTEX_WAIT_BITSET использует битовую маску для более гибкой синхронизации. Это эквивалент pthread_join() на уровне системных вызовов.

Ключевые строки strace логов к лабораторной работе 3:

```
openat(AT_FDCWD, "/tmp/shared_memory O_RDWR|O_CREAT|O_TRUNC, 0666) = 3
ftruncate(3, 264) = 0
mmap(NULL, 264, PROT_READ|PROT_WRITE, MAP_SHARED, 3, 0) = 0x7aa519ef0000
```

Создание общей памяти между всеми процессами (264 байта). Все два процесса (родитель + дочерний) видят одну и ту же физическую память, но в разных виртуальных адресах

```
clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD, child_tidptr=0x7aa519eb37d0) = 23179
```

Создание дочернего процесса

```
openat(AT_FDCWD, "1.txt O_WRONLY|O_CREAT|O_TRUNC, 0666) = 3
```

Открытие файла для записи

```
clock_nanosleep(CLOCK_REALTIME, 0, tv_sec = 0, tv_nsec = 1000000, NULL) = 0
```

Пауза в 1 миллисекунду (1,000,000 наносекунд) между чтениями

Ключевые строки strace логов к лабораторной работе 4:

```
openat(AT_FDCWD, "./libgcd_sin1.so O_RDONLY|O_CLOEXEC) = 3
mmap(NULL, 16416, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7b2d366f4000
mmap(0x7b2d366f5000, 4096, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1000)
mmap(0x7b2d366f6000, 4096, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2000)
mmap(0x7b2d366f7000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2000)
```

```
openat(AT_FDCWD, "./libgcd_sin2.so O_RDONLY|O_CLOEXEC) = 3
mmap(NULL, 16416, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7b2d366f4000
```

Подгрузка библиотек

```
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/glibc-hwcap/x86-64-v3/libgcd_sin1.so ...")
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/glibc-hwcap/x86-64-v2/libgcd_sin1.so ...")
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libgcd_sin1.so ...")
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcd_sin1.so ...")
openat(AT_FDCWD, "./libgcd_sin1.so ...) УСПЕХ!
```

Поиск библиотек по стандартным путям

```
munmap(0x7b2d366f4000, 16416)
```

Выгрузка библиотек

Отличие от статической загрузки память защищается

```
mprotect(0x7cc6a28b9000, 4096, PROT_READ) = 0
```

Путь поиска библиотек

```
openat(AT_FDCWD, "/home/alaineredd/workspace/OSLabs/4/build/lib/libgcd_sin1.so ...) = 3
```

Результаты

Получены 5 файлов содержащие strace логи к лабораторным работам (к лабораторным работам 1-3 по одному файлу, к лабораторной работе 4 приложены 2 файла для каждой тестовой программы program1 и program2).

Выводы

Цели лабораторной работы достигнуты. Я провел диагностику всех программ лабораторных работ посредством анализа strace логов. Все четыре лабораторные работы полностью соответствуют требованиям заданий. Они демонстрируют эволюцию подходов к параллельному программированию в Linux: от классических pipes к многопоточности, к IPC с

изоляция, к модульности и гибкости. Strace логи подтверждают корректность реализации на уровне системных вызовов, синхронизации и управления ресурсами.

Исходная программа

```
16293 execve("./parent",["./parent"],0x7ffc11df8958 /* 49 vars */)
= 0
16293 brk(NULL) = 0x585b7d10f000
16293
mmap(NULL,8192,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_ANONYMOUS,-1,0) =
0x7007edb73000
16293 access("/etc/ld.so.preload",R_OK) = -1 ENOENT (No such file
or directory)
16293 openat(AT_FDCWD,"/etc/ld.so.cache",O_RDONLY|O_CLOEXEC) = 3
16293 fstat(3,{st_mode=S_IFREG|0644,st_size=60107,...}) = 0
16293 mmap(NULL,60107,PROT_READ,MAP_PRIVATE,3,0) = 0x7007edb64000
16293 close(3) = 0
16293
openat(AT_FDCWD,"/lib/x86_64-linux-gnu/libc.so.6",O_RDONLY|O_CLOEXEC) = 3
16293
read(3,"\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\220\243\2\0\0\0\0\0"... ,832)
= 832
16293
pread64(3,"\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"... ,784,64)
= 784
16293 fstat(3,{st_mode=S_IFREG|0755,st_size=2125328,...}) = 0
16293
pread64(3,"\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"... ,784,64)
= 784
16293 mmap(NULL,2170256,PROT_READ,MAP_PRIVATE|MAP_DENYWRITE,3,0) =
0x7007ed800000
16293
mmap(0x7007ed828000,1605632,PROT_READ|PROT_EXEC,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0)
= 0x7007ed828000
16293
mmap(0x7007ed9b0000,323584,PROT_READ,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x1b0000)
= 0x7007ed9b0000
16293
mmap(0x7007ed9ff000,24576,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0)
= 0x7007ed9ff000
16293
mmap(0x7007eda05000,52624,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS,-1,0)
= 0x7007eda05000
16293 close(3) = 0
16293
mmap(NULL,12288,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_ANONYMOUS,-1,0) =
0x7007edb61000
16293 arch_prctl(ARCH_SET_FS,0x7007edb61740) = 0
16293 set_tid_address(0x7007edb61a10) = 16293
16293 set_robust_list(0x7007edb61a20,24) = 0
16293 rseq(0x7007edb62060,0x20,0,0x53053053) = 0
16293 mprotect(0x7007ed9ff000,16384,PROT_READ) = 0
```

```

16293 mprotect(0x585b6e4a6000,4096,PROT_READ) = 0
16293 mprotect(0x7007edbb1000,8192,PROT_READ) = 0
16293
prlimit64(0,RLIMIT_STACK,NULL,{rlim_cur=8192*1024,rlim_max=RLIM64_INFINITY})
= 0
16293 munmap(0x7007edb64000,60107) = 0
16293 fstat(1,{st_mode=S_IFCHR|0620,st_rdev=makedev(0x88,0),...}) =
0
16293 getrandom("\xb8\xe0\x9c\x50\x40\x53\x1d\x33",8,GRND_NONBLOCK)
= 8
16293 brk(NULL) = 0x585b7d10f000
16293 brk(0x585b7d130000) = 0x585b7d130000
16293 fstat(0,{st_mode=S_IFCHR|0620,st_rdev=makedev(0x88,0),...}) =
0
16293
write(1,"\320\222\320\262\320\265\320\264\320\270\321\202\320\265
\320\270\320\274\321\217 \321\204\320\260\320\271\320\273\320\260",32) =
32
16293 read(0,"popa\n",1024) = 5
16293 pipe2([3,4],0) = 0
16293 pipe2([5,6],0) = 0
16293
clone(child_stack=NULL,flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD,child_tid
= 16295
16293 close(3) = 0
16293 close(6 <unfinished ...>
16295 set_robust_list(0x7007edb61a20,24 <unfinished ...>
16293 <... close resumed>) = 0
16293 read(0, <unfinished ...>
16295 <... set_robust_list resumed>) = 0
16295 close(4) = 0
16295 close(5) = 0
16295 dup2(3,0) = 0
16295 close(3) = 0
16295 dup2(6,1) = 1
16295 close(6) = 0
16295 execve("./child",["child","popa"],0x7ffc371e6c18 /* 49 vars
*/) = 0
16295 brk(NULL) = 0x637b85025000
16295
mmap(NULL,8192,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_ANONYMOUS,-1,0) =
0x7e38ad175000
16295 access("/etc/ld.so.preload",R_OK) = -1 ENOENT (No such file
or directory)
16295 openat(AT_FDCWD,"/etc/ld.so.cache",O_RDONLY|O_CLOEXEC) = 3
16295 fstat(3,{st_mode=S_IFREG|0644,st_size=60107,...}) = 0
16295 mmap(NULL,60107,PROT_READ,MAP_PRIVATE,3,0) = 0x7e38ad166000
16295 close(3) = 0

```

```

16295
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
16295
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\220\243\2\0\0\0\0\0"... ,832)
= 832
16295
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"... ,784,64)
= 784
16295 fstat(3,{st_mode=S_IFREG|0755,st_size=2125328,...}) = 0
16295
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"... ,784,64)
= 784
16295 mmap(NULL,2170256,PROT_READ,MAP_PRIVATE|MAP_DENYWRITE,3,0) =
0x7e38ace00000
16295
mmap(0x7e38ace28000,1605632,PROT_READ|PROT_EXEC,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0)
= 0x7e38ace28000
16295
mmap(0x7e38acfb0000,323584,PROT_READ,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x1b0000)
= 0x7e38acfb0000
16295
mmap(0x7e38acfff000,24576,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0)
= 0x7e38acfff000
16295
mmap(0x7e38ad005000,52624,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS,-1,0)
= 0x7e38ad005000
16295 close(3) = 0
16295
mmap(NULL,12288,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_ANONYMOUS,-1,0) =
0x7e38ad163000
16295 arch_prctl(ARCH_SET_FS,0x7e38ad163740) = 0
16295 set_tid_address(0x7e38ad163a10) = 16295
16295 set_robust_list(0x7e38ad163a20,24) = 0
16295 rseq(0x7e38ad164060,0x20,0,0x53053053) = 0
16295 mprotect(0x7e38acfff000,16384,PROT_READ) = 0
16295 mprotect(0x637b537bb000,4096,PROT_READ) = 0
16295 mprotect(0x7e38ad1b3000,8192,PROT_READ) = 0
16295
prlimit64(0,RLIMIT_STACK,NULL,{rlim_cur=8192*1024,rlim_max=RLIM64_INFINITY})
= 0
16295 munmap(0x7e38ad166000,60107) = 0
16295 getrandom("\xce\xcd\x99\x24\x0f\x53\x88\xfd",8,GRND_NONBLOCK)
= 8
16295 brk(NULL) = 0x637b85025000
16295 brk(0x637b85046000) = 0x637b85046000
16295 openat(AT_FDCWD,"popa",O_WRONLY|O_CREAT|O_TRUNC,0666) = 3
16295 fstat(0,{st_mode=S_IFIFO|0600,st_size=0,...}) = 0
16295 read(0, <unfinished ...>

```



```

16293 <... read resumed>"Papa\n",1024) = 5
16293 write(4,"Papa\n",5) = 5
16293 read(0, <unfinished ...>
16295 <... read resumed>"Papa\n",4096) = 5
16295 fstat(3,{st_mode=S_IFREG|0664,st_size=0,...}) = 0
16295 read(0, <unfinished ...>
16293 <... read resumed>"Fo\n",1024) = 3
16293 write(4,"Fo\n",3) = 3
16295 <... read resumed>"Fo\n",4096) = 3
16293 read(0, <unfinished ...>
16295 read(0, <unfinished ...>
16293 <... read resumed>"fo\n",1024) = 3
16293 write(4,"fo\n",3) = 3
16295 <... read resumed>"fo\n",4096) = 3
16293 read(0, <unfinished ...>
16295 fstat(1,{st_mode=S_IFIFO|0600,st_size=0,...}) = 0
16295 read(0, <unfinished ...>
16293 <... read resumed>"shenen\n",1024) = 7
16293 write(4,"shenen\n",7) = 7
16293 read(0, <unfinished ...>
16295 <... read resumed>"shenen\n",4096) = 7
16295 read(0, <unfinished ...>
16293 <... read resumed>"AAaaa\n",1024) = 6
16293 write(4,"AAaaa\n",6) = 6
16295 <... read resumed>"AAaaa\n",4096) = 6
16293 read(0, <unfinished ...>
16295 read(0, <unfinished ...>
16293 <... read resumed>"aaaaaaa\n",1024) = 8
16293 write(4,"aaaaaaa\n",8) = 8
16295 <... read resumed>"aaaaaaa\n",4096) = 8
16293 read(0, <unfinished ...>
16295 read(0, <unfinished ...>
16293 <... read resumed>"",1024) = 0
16293 close(4) = 0
16293 read(5, <unfinished ...>
16295 <... read resumed>"",4096) = 0
16295 write(3,"Papa\nFo\nAAaaa\n",14) = 14
16295 close(3) = 0
16295 write(1,"line fo\n is not valid\nline shene"... ,75
<unfinished ...>
16293 <... read resumed>"line fo\n is not valid\nline
shene"... ,256) = 75
16295 <... write resumed> = 75
16295 exit_group(0 <unfinished ...>
16293 write(1,"line fo\n is not valid\nline shene"... ,75
<unfinished ...>
16295 <... exit_group resumed> = ?
16293 <... write resumed> = 75

```

```

16293 read(5,"",256) = 0
16295 +++ exited with 0 +++
16293 ---SIGCHLD
{si_signo=SIGCHLD,si_code=CLD_EXITED,si_pid=16295,si_uid=1000,si_status=0,si_utime=0,
---
16293 close(5) = 0
16293 wait4(-1,NULL,0,NULL) = 16295
16293 exit_group(0) = ?
16293 +++ exited with 0 +++

```

Листинг 1: *Strace логи 1*

```

execve("./main",["./main","4"],0x7fff7cb4e9f8 /* 49 vars */) = 0
brk(NULL) = 0x5f8de1c23000
mmap(NULL,8192,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_ANONYMOUS,-1,0)
= 0x7dab8d622000
access("/etc/ld.so.preload",R_OK) = -1 ENOENT (No such file or
directory)
openat(AT_FDCWD,"/etc/ld.so.cache",O_RDONLY|O_CLOEXEC) = 3
fstat(3,{st_mode=S_IFREG|0644,st_size=60107,...}) = 0
mmap(NULL,60107,PROT_READ,MAP_PRIVATE,3,0) = 0x7dab8d613000
close(3) = 0

openat(AT_FDCWD,"/lib/x86_64-linux-gnu/libstdc++.so.6",O_RDONLY|O_CLOEXEC)
= 3

read(3,"\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"... ,832)
= 832
fstat(3,{st_mode=S_IFREG|0644,st_size=2592224,...}) = 0
mmap(NULL,2609472,PROT_READ,MAP_PRIVATE|MAP_DENYWRITE,3,0) =
0x7dab8d200000

mmap(0x7dab8d29d000,1343488,PROT_READ|PROT_EXEC,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,
= 0x7dab8d29d000

mmap(0x7dab8d3e5000,552960,PROT_READ,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x1e5000)
= 0x7dab8d3e5000

mmap(0x7dab8d46c000,57344,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,
= 0x7dab8d46c000

mmap(0x7dab8d47a000,12608,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS,-1
= 0x7dab8d47a000
close(3) = 0

openat(AT_FDCWD,"/lib/x86_64-linux-gnu/libgcc_s.so.1",O_RDONLY|O_CLOEXEC)
= 3

```

```

read(3,"\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"... ,832)
= 832
    fstat(3,{st_mode=S_IFREG|0644,st_size=183024,...}) = 0
    mmap(NULL,185256,PROT_READ,MAP_PRIVATE|MAP_DENYWRITE,3,0) =
0x7dab8d5e5000

mmap(0x7dab8d5e9000,147456,PROT_READ|PROT_EXEC,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0)
= 0x7dab8d5e9000

mmap(0x7dab8d60d000,16384,PROT_READ,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x28000)
= 0x7dab8d60d000

mmap(0x7dab8d611000,8192,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0)
= 0x7dab8d611000
    close(3) = 0

openat(AT_FDCWD,"/lib/x86_64-linux-gnu/libc.so.6",O_RDONLY|O_CLOEXEC) = 3

read(3,"\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\220\243\2\0\0\0\0\0"... ,832)
= 832

pread64(3,"\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"... ,784,64)
= 784
    fstat(3,{st_mode=S_IFREG|0755,st_size=2125328,...}) = 0

pread64(3,"\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"... ,784,64)
= 784
    mmap(NULL,2170256,PROT_READ,MAP_PRIVATE|MAP_DENYWRITE,3,0) =
0x7dab8ce00000

mmap(0x7dab8ce28000,1605632,PROT_READ|PROT_EXEC,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0)
= 0x7dab8ce28000

mmap(0x7dab8cfb0000,323584,PROT_READ,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x1b0000)
= 0x7dab8cfb0000

mmap(0x7dab8cfff000,24576,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0)
= 0x7dab8cfff000

mmap(0x7dab8d005000,52624,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS,-1,0)
= 0x7dab8d005000
    close(3) = 0

openat(AT_FDCWD,"/lib/x86_64-linux-gnu/libm.so.6",O_RDONLY|O_CLOEXEC) = 3

read(3,"\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"... ,832)
= 832

```

```

    fstat(3,{st_mode=S_IFREG|0644,st_size=952616,...}) = 0
    mmap(NULL,950296,PROT_READ,MAP_PRIVATE|MAP_DENYWRITE,3,0) =
0x7dab8d4fc000

mmap(0x7dab8d50c000,520192,PROT_READ|PROT_EXEC,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0) =
= 0x7dab8d50c000

mmap(0x7dab8d58b000,360448,PROT_READ,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x8f000)
= 0x7dab8d58b000

mmap(0x7dab8d5e3000,8192,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0)
= 0x7dab8d5e3000
    close(3) = 0
    mmap(NULL,8192,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_ANONYMOUS,-1,0)
= 0x7dab8d4fa000

mmap(NULL,12288,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_ANONYMOUS,-1,0) =
0x7dab8d4f7000
    arch_prctl(ARCH_SET_FS,0x7dab8d4f7740) = 0
    set_tid_address(0x7dab8d4f7a10) = 23409
    set_robust_list(0x7dab8d4f7a20,24) = 0
    rseq(0x7dab8d4f8060,0x20,0,0x53053053) = 0
    mprotect(0x7dab8cfff000,16384,PROT_READ) = 0
    mprotect(0x7dab8d5e3000,4096,PROT_READ) = 0
    mprotect(0x7dab8d611000,4096,PROT_READ) = 0
    mprotect(0x7dab8d46c000,45056,PROT_READ) = 0
    mprotect(0x5f8dcb8bb000,4096,PROT_READ) = 0
    mprotect(0x7dab8d660000,8192,PROT_READ) = 0

prlimit64(0,RLIMIT_STACK,NULL,{rlim_cur=8192*1024,rlim_max=RLIM64_INFINITY})
= 0
    munmap(0x7dab8d613000,60107) = 0
    futex(0x7dab8d47a7bc,FUTEX_WAKE_PRIVATE,2147483647) = 0
    getrandom("\x20\x47\x0d\x4e\xb6\xdf\x18\xc9",8,GRND_NONBLOCK) = 8
    brk(NULL) = 0x5f8de1c23000
    brk(0x5f8de1c44000) = 0x5f8de1c44000
    fstat(1,{st_mode=S_IFCHR|0620,st_rdev=makedev(0x88,0),...}) = 0
    write(1,"\320\222\320\262\320\265\320\264\320\270\321\202\320\265
\321\200\320\260\320\267\320\274\320\265\321\200\321\213 \320\274"... ,48)
= 48
    fstat(0,{st_mode=S_IFCHR|0620,st_rdev=makedev(0x88,0),...}) = 0
    read(0,"500\n",1024) = 4
    read(0,"500\n",1024) = 4
    write(1,"\320\262\320\262\320\265\320\264\320\270\321\202\320\265
\321\200\320\260\320\267\320\274\320\265\321\200\321\213 \320\274"... ,48)
= 48
    read(0,"500\n",1024) = 4
    read(0,"500\n",1024) = 4

```

brk(0x5f8de1c65000)	= 0x5f8de1c65000
brk(0x5f8de1c86000)	= 0x5f8de1c86000
brk(0x5f8de1ca7000)	= 0x5f8de1ca7000
brk(0x5f8de1cc9000)	= 0x5f8de1cc9000
brk(0x5f8de1cea000)	= 0x5f8de1cea000
brk(0x5f8de1d0b000)	= 0x5f8de1d0b000
brk(0x5f8de1d2c000)	= 0x5f8de1d2c000
brk(0x5f8de1d4e000)	= 0x5f8de1d4e000
brk(0x5f8de1d6f000)	= 0x5f8de1d6f000
brk(0x5f8de1d90000)	= 0x5f8de1d90000
brk(0x5f8de1db2000)	= 0x5f8de1db2000
brk(0x5f8de1dd3000)	= 0x5f8de1dd3000
brk(0x5f8de1df4000)	= 0x5f8de1df4000
brk(0x5f8de1e15000)	= 0x5f8de1e15000
brk(0x5f8de1e37000)	= 0x5f8de1e37000
brk(0x5f8de1e58000)	= 0x5f8de1e58000
brk(0x5f8de1e79000)	= 0x5f8de1e79000
brk(0x5f8de1e9a000)	= 0x5f8de1e9a000
brk(0x5f8de1ebc000)	= 0x5f8de1ebc000
brk(0x5f8de1edd000)	= 0x5f8de1edd000
brk(0x5f8de1efe000)	= 0x5f8de1efe000
brk(0x5f8de1f20000)	= 0x5f8de1f20000
brk(0x5f8de1f41000)	= 0x5f8de1f41000
brk(0x5f8de1f62000)	= 0x5f8de1f62000
brk(0x5f8de1f83000)	= 0x5f8de1f83000
brk(0x5f8de1fa5000)	= 0x5f8de1fa5000
brk(0x5f8de1fc6000)	= 0x5f8de1fc6000
brk(0x5f8de1fe7000)	= 0x5f8de1fe7000
brk(0x5f8de2008000)	= 0x5f8de2008000
brk(0x5f8de202a000)	= 0x5f8de202a000
brk(0x5f8de204c000)	= 0x5f8de204c000
brk(0x5f8de206d000)	= 0x5f8de206d000
brk(0x5f8de208e000)	= 0x5f8de208e000
brk(0x5f8de20b0000)	= 0x5f8de20b0000
brk(0x5f8de20d1000)	= 0x5f8de20d1000
brk(0x5f8de20f2000)	= 0x5f8de20f2000
brk(0x5f8de2114000)	= 0x5f8de2114000
brk(0x5f8de2135000)	= 0x5f8de2135000
brk(0x5f8de2156000)	= 0x5f8de2156000
brk(0x5f8de2177000)	= 0x5f8de2177000
brk(0x5f8de2199000)	= 0x5f8de2199000
brk(0x5f8de21ba000)	= 0x5f8de21ba000
brk(0x5f8de21db000)	= 0x5f8de21db000
brk(0x5f8de21fc000)	= 0x5f8de21fc000
brk(0x5f8de221e000)	= 0x5f8de221e000
brk(0x5f8de223f000)	= 0x5f8de223f000
brk(0x5f8de2260000)	= 0x5f8de2260000
brk(0x5f8de2281000)	= 0x5f8de2281000

```

brk(0x5f8de22a3000)      = 0x5f8de22a3000
brk(0x5f8de22c4000)      = 0x5f8de22c4000
brk(0x5f8de22e5000)      = 0x5f8de22e5000
brk(0x5f8de2307000)      = 0x5f8de2307000
brk(0x5f8de2328000)      = 0x5f8de2328000
brk(0x5f8de2349000)      = 0x5f8de2349000
brk(0x5f8de236a000)      = 0x5f8de236a000
brk(0x5f8de238c000)      = 0x5f8de238c000
brk(0x5f8de23ad000)      = 0x5f8de23ad000
brk(0x5f8de23ce000)      = 0x5f8de23ce000
brk(0x5f8de23ef000)      = 0x5f8de23ef000
brk(0x5f8de2410000)      = 0x5f8de2410000
brk(0x5f8de2431000)      = 0x5f8de2431000
brk(0x5f8de2452000)      = 0x5f8de2452000
brk(0x5f8de2474000)      = 0x5f8de2474000
brk(0x5f8de2495000)      = 0x5f8de2495000
brk(0x5f8de24b6000)      = 0x5f8de24b6000
brk(0x5f8de24d7000)      = 0x5f8de24d7000
brk(0x5f8de24f9000)      = 0x5f8de24f9000
brk(0x5f8de251a000)      = 0x5f8de251a000
brk(0x5f8de253b000)      = 0x5f8de253b000
brk(0x5f8de255c000)      = 0x5f8de255c000
brk(0x5f8de257e000)      = 0x5f8de257e000
brk(0x5f8de259f000)      = 0x5f8de259f000
brk(0x5f8de25c0000)      = 0x5f8de25c0000
brk(0x5f8de25e2000)      = 0x5f8de25e2000
brk(0x5f8de2603000)      = 0x5f8de2603000
brk(0x5f8de2624000)      = 0x5f8de2624000
brk(0x5f8de2645000)      = 0x5f8de2645000
brk(0x5f8de2667000)      = 0x5f8de2667000
brk(0x5f8de2688000)      = 0x5f8de2688000
brk(0x5f8de26a9000)      = 0x5f8de26a9000
brk(0x5f8de26ca000)      = 0x5f8de26ca000
brk(0x5f8de26ec000)      = 0x5f8de26ec000
brk(0x5f8de270d000)      = 0x5f8de270d000
brk(0x5f8de272e000)      = 0x5f8de272e000
brk(0x5f8de274f000)      = 0x5f8de274f000
brk(0x5f8de2771000)      = 0x5f8de2771000
brk(0x5f8de2792000)      = 0x5f8de2792000
brk(0x5f8de27b3000)      = 0x5f8de27b3000
brk(0x5f8de27d5000)      = 0x5f8de27d5000

```

```

rt_sigaction(SIGRT_1,{sa_handler=0x7dab8ce99530,sa_mask=[],sa_flags=SA_RESTORER|SA_ONSTACK,
= 0

```

```

    rt_sigprocmask(SIG_UNBLOCK,[RTMIN RT_1],NULL,8) = 0

```

```

mmap(NULL,8392704,PROT_NONE,MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK,-1,0) =
0x7dab8c5ff000

```

```

mprotect(0x7dab8c600000,8388608,PROT_READ|PROT_WRITE) = 0
rt_sigprocmask(SIG_BLOCK,~[],[],8) = 0

clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|
=>{parent_tid=[23410]},88) = 23410
rt_sigprocmask(SIG_SETMASK,[],NULL,8) = 0

mmap(NULL,8392704,PROT_NONE,MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK,-1,0) =
0x7dab8bdfef000
mprotect(0x7dab8bdff000,8388608,PROT_READ|PROT_WRITE) = 0
rt_sigprocmask(SIG_BLOCK,~[],[],8) = 0

clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|
=>{parent_tid=[23411]},88) = 23411
rt_sigprocmask(SIG_SETMASK,[],NULL,8) = 0

mmap(NULL,8392704,PROT_NONE,MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK,-1,0) =
0x7dab8b5fd000
mprotect(0x7dab8b5fe000,8388608,PROT_READ|PROT_WRITE) = 0
rt_sigprocmask(SIG_BLOCK,~[],[],8) = 0

clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|
=>{parent_tid=[23412]},88) = 23412
rt_sigprocmask(SIG_SETMASK,[],NULL,8) = 0

mmap(NULL,8392704,PROT_NONE,MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK,-1,0) =
0x7dab8adfc000
mprotect(0x7dab8adfd000,8388608,PROT_READ|PROT_WRITE) = 0
rt_sigprocmask(SIG_BLOCK,~[],[],8) = 0

clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|
=>{parent_tid=[23413]},88) = 23413
rt_sigprocmask(SIG_SETMASK,[],NULL,8) = 0

futex(0x7dab8cdf990,FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME,23410,NULL,FUTEX_BITSET_M
= 0

futex(0x7dab8c5fe990,FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME,23411,NULL,FUTEX_BITSET_M
= 0

write(1,"488326ms\n",9) = 9
brk(0x5f8de2406000) = 0x5f8de2406000
brk(0x5f8de2403000) = 0x5f8de2403000
brk(0x5f8de2031000) = 0x5f8de2031000
brk(0x5f8de202e000) = 0x5f8de202e000
brk(0x5f8de1c5b000) = 0x5f8de1c5b000
brk(0x5f8de1c57000) = 0x5f8de1c57000
lseek(0,-1,SEEK_CUR) = -1 ESPIPE (Illegal seek)
exit_group(0) = ?

```

+++ exited with 0 +++

Листинг 2: *Strace логи 2*

```
execve("./parent",["./parent"],0x7ffceb4d0c00 /* 49 vars */) = 0
brk(NULL) = 0x5d7d814ea000
mmap(NULL,8192,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_ANONYMOUS,-1,0)
= 0x7aa519ef1000
access("/etc/ld.so.preload",R_OK) = -1 ENOENT (No such file or
directory)
openat(AT_FDCWD,"/etc/ld.so.cache",O_RDONLY|O_CLOEXEC) = 3
fstat(3,{st_mode=S_IFREG|0644,st_size=60107,...}) = 0
mmap(NULL,60107,PROT_READ,MAP_PRIVATE,3,0) = 0x7aa519ee2000
close(3) = 0

openat(AT_FDCWD,"/lib/x86_64-linux-gnu/libstdc++.so.6",O_RDONLY|O_CLOEXEC)
= 3

read(3,"\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"... ,832)
= 832
fstat(3,{st_mode=S_IFREG|0644,st_size=2592224,...}) = 0
mmap(NULL,2609472,PROT_READ,MAP_PRIVATE|MAP_DENYWRITE,3,0) =
0x7aa519c00000

mmap(0x7aa519c9d000,1343488,PROT_READ|PROT_EXEC,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,
= 0x7aa519c9d000

mmap(0x7aa519de5000,552960,PROT_READ,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x1e5000)
= 0x7aa519de5000

mmap(0x7aa519e6c000,57344,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,
= 0x7aa519e6c000

mmap(0x7aa519e7a000,12608,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS,-1
= 0x7aa519e7a000
close(3) = 0

openat(AT_FDCWD,"/lib/x86_64-linux-gnu/libc.so.6",O_RDONLY|O_CLOEXEC) = 3

read(3,"\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\220\243\2\0\0\0\0\0"... ,832)
= 832

pread64(3,"\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"... ,784,64)
= 784
fstat(3,{st_mode=S_IFREG|0755,st_size=2125328,...}) = 0

pread64(3,"\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"... ,784,64)
= 784
```



```

mmap(NULL,2170256,PROT_READ,MAP_PRIVATE|MAP_DENYWRITE,3,0) =
0x7aa519800000

mmap(0x7aa519828000,1605632,PROT_READ|PROT_EXEC,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0) =
0x7aa519828000

mmap(0x7aa5199b0000,323584,PROT_READ,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x1b0000) =
0x7aa5199b0000

mmap(0x7aa5199ff000,24576,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0) =
0x7aa5199ff000

mmap(0x7aa519a05000,52624,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS,-1,0) =
0x7aa519a05000
close(3) = 0

openat(AT_FDCWD,"/lib/x86_64-linux-gnu/libm.so.6",O_RDONLY|O_CLOEXEC) = 3

read(3,"\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"... ,832)
= 832
fstat(3,{st_mode=S_IFREG|0644,st_size=952616,...}) = 0
mmap(NULL,950296,PROT_READ,MAP_PRIVATE|MAP_DENYWRITE,3,0) =
0x7aa519b17000

mmap(0x7aa519b27000,520192,PROT_READ|PROT_EXEC,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0) =
0x7aa519b27000

mmap(0x7aa519ba6000,360448,PROT_READ,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x8f000) =
0x7aa519ba6000

mmap(0x7aa519bfe000,8192,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0) =
0x7aa519bfe000
close(3) = 0

openat(AT_FDCWD,"/lib/x86_64-linux-gnu/libgcc_s.so.1",O_RDONLY|O_CLOEXEC)
= 3

read(3,"\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"... ,832)
= 832
fstat(3,{st_mode=S_IFREG|0644,st_size=183024,...}) = 0
mmap(NULL,185256,PROT_READ,MAP_PRIVATE|MAP_DENYWRITE,3,0) =
0x7aa519eb4000

mmap(0x7aa519eb8000,147456,PROT_READ|PROT_EXEC,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0) =
0x7aa519eb8000

mmap(0x7aa519edc000,16384,PROT_READ,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x28000) =
0x7aa519edc000

```

```

mmap(0x7aa519ee0000,8192,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0)
= 0x7aa519ee0000
    close(3) = 0
    mmap(NULL,8192,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_ANONYMOUS,-1,0)
= 0x7aa519eb2000
    arch_prctl(ARCH_SET_FS,0x7aa519eb3500) = 0
    set_tid_address(0x7aa519eb37d0) = 23177
    set_robust_list(0x7aa519eb37e0,24) = 0
    rseq(0x7aa519eb3e20,0x20,0,0x53053053) = 0
    mprotect(0x7aa5199ff000,16384,PROT_READ) = 0
    mprotect(0x7aa519ee0000,4096,PROT_READ) = 0
    mprotect(0x7aa519bfe000,4096,PROT_READ) = 0
    mmap(NULL,8192,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_ANONYMOUS,-1,0)
= 0x7aa519eb0000
    mprotect(0x7aa519e6c000,45056,PROT_READ) = 0
    mprotect(0x5d7d653ee000,4096,PROT_READ) = 0
    mprotect(0x7aa519f2f000,8192,PROT_READ) = 0

prlimit64(0,RLIMIT_STACK,NULL,{rlim_cur=8192*1024,rlim_max=RLIM64_INFINITY})
= 0
    munmap(0x7aa519ee2000,60107) = 0
    futex(0x7aa519e7a7bc,FUTEX_WAKE_PRIVATE,2147483647) = 0
    getrandom("\x32\x69\x36\xc9\x76\xc8\x56\xe1",8,GRND_NONBLOCK) = 8
    brk(NULL) = 0x5d7d814ea000
    brk(0x5d7d8150b000) = 0x5d7d8150b000
    fstat(1,{st_mode=S_IFCHR|0620,st_rdev=makedev(0x88,0),...}) = 0
    fstat(0,{st_mode=S_IFCHR|0620,st_rdev=makedev(0x88,0),...}) = 0
    write(1,"\320\222\320\262\320\265\320\264\320\270\321\202\320\265
\320\270\320\274\321\217 \321\204\320\260\320\271\320\273\320\260"... ,64)
= 64
    read(0,"1.txt\n",1024) = 6
    openat(AT_FDCWD,"/tmp/shared_memory",O_RDWR|O_CREAT|O_TRUNC,0666) =
3
    ftruncate(3,264) = 0
    mmap(NULL,264,PROT_READ|PROT_WRITE,MAP_SHARED,3,0) = 0x7aa519ef0000
    close(3) = 0

clone(child_stack=NULL,flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD,child_tid
= 23179
    write(1,"parent proc\n",12) = 12
    openat(AT_FDCWD,"1.txt",O_WRONLY|O_CREAT|O_TRUNC,0666) = 3
    read(0,"dasd\n",1024) = 5
    clock_nanosleep(CLOCK_REALTIME,0,{tv_sec=0,tv_nsec=1000000},NULL) =
0
    read(0,"Das\n",1024) = 4
    clock_nanosleep(CLOCK_REALTIME,0,{tv_sec=0,tv_nsec=1000000},NULL) =
0

```

```

read(0,"Daaaaa\n",1024)                = 7
clock_nanosleep(CLOCK_REALTIME,0,{tv_sec=0,tv_nsec=1000000},NULL) =
0
read(0,"sss\n",1024)                    = 4
clock_nanosleep(CLOCK_REALTIME,0,{tv_sec=0,tv_nsec=1000000},NULL) =
0
read(0,"0\n",1024)                      = 2
clock_nanosleep(CLOCK_REALTIME,0,{tv_sec=0,tv_nsec=1000000},NULL) =
0
read(0,"1\n",1024)                      = 2
clock_nanosleep(CLOCK_REALTIME,0,{tv_sec=0,tv_nsec=1000000},NULL) =
0
read(0,"\n",1024)                       = 1
clock_nanosleep(CLOCK_REALTIME,0,{tv_sec=0,tv_nsec=1000000},NULL) =
0
read(0,0x5d7d814fc6c0,1024)            = ? ERESTARTSYS (To be
restarted if SA_RESTART is set)
---SIGINT {si_signo=SIGINT,si_code=SI_KERNEL} ---
+++ killed by SIGINT +++

```

Листинг 3: *Strace логи 3*

```

execve("./program2",["./program2"],0x7fffeef88620 /* 49 vars */) =
0
brk(NULL)                                = 0x64f595835000
mmap(NULL,8192,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_ANONYMOUS,-1,0)
= 0x7b2d366f9000
access("/etc/ld.so.preload",R_OK)        = -1 ENOENT (No such file or
directory)
openat(AT_FDCWD,"/etc/ld.so.cache",O_RDONLY|O_CLOEXEC) = 3
fstat(3,{st_mode=S_IFREG|0644,st_size=60107,...}) = 0
mmap(NULL,60107,PROT_READ,MAP_PRIVATE,3,0) = 0x7b2d366ea000
close(3)                                 = 0

openat(AT_FDCWD,"/lib/x86_64-linux-gnu/libstdc++.so.6",O_RDONLY|O_CLOEXEC)
= 3

read(3,"\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"... ,832)
= 832
fstat(3,{st_mode=S_IFREG|0644,st_size=2592224,...}) = 0
mmap(NULL,2609472,PROT_READ,MAP_PRIVATE|MAP_DENYWRITE,3,0) =
0x7b2d36400000

mmap(0x7b2d3649d000,1343488,PROT_READ|PROT_EXEC,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3
= 0x7b2d3649d000

mmap(0x7b2d365e5000,552960,PROT_READ,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x1e5000)
= 0x7b2d365e5000

```

```

mmap(0x7b2d3666c000,57344,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,
= 0x7b2d3666c000

mmap(0x7b2d3667a000,12608,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS,-1
= 0x7b2d3667a000
    close(3)                                = 0

openat(AT_FDCWD,"/lib/x86_64-linux-gnu/libgcc_s.so.1",O_RDONLY|O_CLOEXEC)
= 3

read(3,"\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"...,832)
= 832
    fstat(3,{st_mode=S_IFREG|0644,st_size=183024,...}) = 0
    mmap(NULL,185256,PROT_READ,MAP_PRIVATE|MAP_DENYWRITE,3,0) =
0x7b2d366bc000

mmap(0x7b2d366c0000,147456,PROT_READ|PROT_EXEC,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,
= 0x7b2d366c0000

mmap(0x7b2d366e4000,16384,PROT_READ,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x28000)
= 0x7b2d366e4000

mmap(0x7b2d366e8000,8192,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0
= 0x7b2d366e8000
    close(3)                                = 0

openat(AT_FDCWD,"/lib/x86_64-linux-gnu/libc.so.6",O_RDONLY|O_CLOEXEC) = 3

read(3,"\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\220\243\2\0\0\0\0\0"...,832)
= 832

pread64(3,"\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"...,784,64)
= 784
    fstat(3,{st_mode=S_IFREG|0755,st_size=2125328,...}) = 0

pread64(3,"\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"...,784,64)
= 784
    mmap(NULL,2170256,PROT_READ,MAP_PRIVATE|MAP_DENYWRITE,3,0) =
0x7b2d36000000

mmap(0x7b2d36028000,1605632,PROT_READ|PROT_EXEC,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,
= 0x7b2d36028000

mmap(0x7b2d361b0000,323584,PROT_READ,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x1b0000)
= 0x7b2d361b0000

mmap(0x7b2d361ff000,24576,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,

```

```

= 0x7b2d361ff000

mmap(0x7b2d36205000,52624,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS,-1,0) = 0x7b2d36205000
close(3) = 0

openat(AT_FDCWD,"/lib/x86_64-linux-gnu/libm.so.6",O_RDONLY|O_CLOEXEC) = 3

read(3,"\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"... ,832)
= 832
fstat(3,{st_mode=S_IFREG|0644,st_size=952616,...}) = 0
mmap(NULL,950296,PROT_READ,MAP_PRIVATE|MAP_DENYWRITE,3,0) =
0x7b2d36317000

mmap(0x7b2d36327000,520192,PROT_READ|PROT_EXEC,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0) = 0x7b2d36327000

mmap(0x7b2d363a6000,360448,PROT_READ,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x8f000) = 0x7b2d363a6000

mmap(0x7b2d363fe000,8192,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0) = 0x7b2d363fe000
close(3) = 0
mmap(NULL,8192,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_ANONYMOUS,-1,0) = 0x7b2d366ba000

mmap(NULL,12288,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_ANONYMOUS,-1,0) =
0x7b2d366b7000
arch_prctl(ARCH_SET_FS,0x7b2d366b7740) = 0
set_tid_address(0x7b2d366b7a10) = 15880
set_robust_list(0x7b2d366b7a20,24) = 0
rseq(0x7b2d366b8060,0x20,0,0x53053053) = 0
mprotect(0x7b2d361ff000,16384,PROT_READ) = 0
mprotect(0x7b2d363fe000,4096,PROT_READ) = 0
mprotect(0x7b2d366e8000,4096,PROT_READ) = 0
mprotect(0x7b2d3666c000,45056,PROT_READ) = 0
mprotect(0x64f582cfe000,4096,PROT_READ) = 0
mprotect(0x7b2d36737000,8192,PROT_READ) = 0

prlimit64(0,RLIMIT_STACK,NULL,{rlim_cur=8192*1024,rlim_max=RLIM64_INFINITY})
= 0
munmap(0x7b2d366ea000,60107) = 0
futex(0x7b2d3667a7bc,FUTEX_WAKE_PRIVATE,2147483647) = 0
getrandom("\xa7\x0b\xe7\x10\x7c\xe5\x15\x55",8,GRND_NONBLOCK) = 8
brk(NULL) = 0x64f595835000
brk(0x64f595856000) = 0x64f595856000
fstat(1,{st_mode=S_IFCHR|0620,st_rdev=makedev(0x88,0x1),...}) = 0

```

```

write(1, "\320\227\320\260\320\263\321\200\321\203\320\266\320\260\320\265\320\274
\320\261\320\270\320\261\320\273\320\270\320\276\321"... ,56) = 56
    openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
    fstat(3, {st_mode=S_IFREG|0644, st_size=60107, ...}) = 0
    mmap(NULL, 60107, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7b2d366ea000
    close(3) = 0

openat(AT_FDCWD, "/lib/x86_64-linux-gnu/glibc-hwcaps/x86-64-v3/libgcd_sin1.so", O_RDONLY)
= -1 ENOENT (No such file or directory)

newfstatat(AT_FDCWD, "/lib/x86_64-linux-gnu/glibc-hwcaps/x86-64-v3/", 0x7fff1483fe40, 0)
= -1 ENOENT (No such file or directory)

openat(AT_FDCWD, "/lib/x86_64-linux-gnu/glibc-hwcaps/x86-64-v2/libgcd_sin1.so", O_RDONLY)
= -1 ENOENT (No such file or directory)

newfstatat(AT_FDCWD, "/lib/x86_64-linux-gnu/glibc-hwcaps/x86-64-v2/", 0x7fff1483fe40, 0)
= -1 ENOENT (No such file or directory)

openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libgcd_sin1.so", O_RDONLY|O_CLOEXEC)
= -1 ENOENT (No such file or directory)

newfstatat(AT_FDCWD, "/lib/x86_64-linux-gnu/", {st_mode=S_IFDIR|0755, st_size=86016, ...})
= 0

openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/glibc-hwcaps/x86-64-v3/libgcd_sin1.so", O_RDONLY)
= -1 ENOENT (No such file or directory)

newfstatat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/glibc-hwcaps/x86-64-v3/", 0x7fff1483fe40, 0)
= -1 ENOENT (No such file or directory)

openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/glibc-hwcaps/x86-64-v2/libgcd_sin1.so", O_RDONLY)
= -1 ENOENT (No such file or directory)

newfstatat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/glibc-hwcaps/x86-64-v2/", 0x7fff1483fe40, 0)
= -1 ENOENT (No such file or directory)

openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcd_sin1.so", O_RDONLY|O_CLOEXEC)
= -1 ENOENT (No such file or directory)

newfstatat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/", {st_mode=S_IFDIR|0755, st_size=86016, ...})
= 0

openat(AT_FDCWD, "/lib/glibc-hwcaps/x86-64-v3/libgcd_sin1.so", O_RDONLY|O_CLOEXEC)
= -1 ENOENT (No such file or directory)

newfstatat(AT_FDCWD, "/lib/glibc-hwcaps/x86-64-v3/", 0x7fff1483fe40, 0) = -1
ENOENT (No such file or directory)

```

```

openat(AT_FDCWD, "/lib/glibc-hwcaps/x86-64-v2/libgcd_sin1.so", O_RDONLY|O_CLOEXEC)
= -1 ENOENT (No such file or directory)

newfstatat(AT_FDCWD, "/lib/glibc-hwcaps/x86-64-v2/", 0x7fff1483fe40, 0) = -1
ENOENT (No such file or directory)
    openat(AT_FDCWD, "/lib/libgcd_sin1.so", O_RDONLY|O_CLOEXEC) = -1
ENOENT (No such file or directory)

newfstatat(AT_FDCWD, "/lib/", {st_mode=S_IFDIR|0755, st_size=4096, ...}, 0) =
0

openat(AT_FDCWD, "/usr/lib/glibc-hwcaps/x86-64-v3/libgcd_sin1.so", O_RDONLY|O_CLOEXEC)
= -1 ENOENT (No such file or directory)

newfstatat(AT_FDCWD, "/usr/lib/glibc-hwcaps/x86-64-v3/", 0x7fff1483fe40, 0)
= -1 ENOENT (No such file or directory)

openat(AT_FDCWD, "/usr/lib/glibc-hwcaps/x86-64-v2/libgcd_sin1.so", O_RDONLY|O_CLOEXEC)
= -1 ENOENT (No such file or directory)

newfstatat(AT_FDCWD, "/usr/lib/glibc-hwcaps/x86-64-v2/", 0x7fff1483fe40, 0)
= -1 ENOENT (No such file or directory)
    openat(AT_FDCWD, "/usr/lib/libgcd_sin1.so", O_RDONLY|O_CLOEXEC) = -1
ENOENT (No such file or directory)

newfstatat(AT_FDCWD, "/usr/lib/", {st_mode=S_IFDIR|0755, st_size=4096, ...}, 0)
= 0
    munmap(0x7b2d366ea000, 60107) = 0
    openat(AT_FDCWD, "./libgcd_sin1.so", O_RDONLY|O_CLOEXEC) = 3

read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"..., 832)
= 832
    fstat(3, {st_mode=S_IFREG|0775, st_size=15536, ...}) = 0
    getcwd("/home/alaineredd/workspace/OSLabs/4/build/bin", 128) = 46
    mmap(NULL, 16416, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) =
0x7b2d366f4000

mmap(0x7b2d366f5000, 4096, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x
= 0x7b2d366f5000

mmap(0x7b2d366f6000, 4096, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2000)
= 0x7b2d366f6000

mmap(0x7b2d366f7000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0
= 0x7b2d366f7000
    close(3) = 0
    mprotect(0x7b2d366f7000, 4096, PROT_READ) = 0

```

```

write(1,"\\320\\221\\320\\270\\320\\261\\320\\273\\320\\270\\320\\276\\321\\202\\320\\265\\320\\272\\320\\
\\321\\203\\321\\201\\320\\277\\320\\265\\321\\210\\320"... ,73) = 73
    write(1,">",2) = 2
    fstat(0,{st_mode=S_IFCHR|0620,st_rdev=makedev(0x88,0x1),...}) = 0
    read(0,"1 0 3.14159 0.001\\n",1024) = 18
    write(1,"\\342\\210\\253sin(x)dx \\320\\276\\321\\202 0 \\320\\264\\320\\276
3.14159 "... ,56) = 56
    write(1,"\\320\\234\\320\\265\\321\\202\\320\\276\\320\\264:
\\320\\277\\321\\200\\321\\217\\320\\274\\320\\276\\321\\203\\320\\263\\320\\276\\320\\273\\321\\214"... ,
= 41
    write(1,">",2) = 2
    read(0,"0\\n",1024) = 2
    write(1,"lib1->lib2\\n",11) = 11
    munmap(0x7b2d366f4000,16416) = 0

write(1,"\\320\\227\\320\\260\\320\\263\\321\\200\\321\\203\\320\\266\\320\\260\\320\\265\\320\\274
\\320\\261\\320\\270\\320\\261\\320\\273\\320\\270\\320\\276\\321"... ,56) = 56
    openat(AT_FDCWD,"/etc/ld.so.cache",O_RDONLY|O_CLOEXEC) = 3
    fstat(3,{st_mode=S_IFREG|0644,st_size=60107,...}) = 0
    mmap(NULL,60107,PROT_READ,MAP_PRIVATE,3,0) = 0x7b2d366ea000
    close(3) = 0

openat(AT_FDCWD,"/lib/x86_64-linux-gnu/libgcd_sin2.so",O_RDONLY|O_CLOEXEC)
= -1 ENOENT (No such file or directory)

openat(AT_FDCWD,"/usr/lib/x86_64-linux-gnu/libgcd_sin2.so",O_RDONLY|O_CLOEXEC)
= -1 ENOENT (No such file or directory)
    openat(AT_FDCWD,"/lib/libgcd_sin2.so",O_RDONLY|O_CLOEXEC) = -1
ENOENT (No such file or directory)
    openat(AT_FDCWD,"/usr/lib/libgcd_sin2.so",O_RDONLY|O_CLOEXEC) = -1
ENOENT (No such file or directory)
    munmap(0x7b2d366ea000,60107) = 0
    openat(AT_FDCWD,"./libgcd_sin2.so",O_RDONLY|O_CLOEXEC) = 3

read(3,"\\177ELF\\2\\1\\1\\0\\0\\0\\0\\0\\0\\0\\0\\0\\0\\3\\0>\\0\\1\\0\\0\\0\\0\\0\\0\\0\\0\\0\\0"... ,832)
= 832
    fstat(3,{st_mode=S_IFREG|0775,st_size=15608,...}) = 0
    getcwd("/home/alaineredd/workspace/OSLabs/4/build/bin",128) = 46
    mmap(NULL,16416,PROT_READ,MAP_PRIVATE|MAP_DENYWRITE,3,0) =
0x7b2d366f4000

mmap(0x7b2d366f5000,4096,PROT_READ|PROT_EXEC,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x
= 0x7b2d366f5000

mmap(0x7b2d366f6000,4096,PROT_READ,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x2000)
= 0x7b2d366f6000

```



```

mmap(0x7b2d366f7000,8192,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0)
= 0x7b2d366f7000
close(3) = 0
mprotect(0x7b2d366f7000,4096,PROT_READ) = 0

write(1,"\320\221\320\270\320\261\320\273\320\270\320\276\321\202\320\265\320\272\320\
\321\203\321\201\320\277\320\265\321\210\320"... ,73) = 73
write(1,">",2) = 2
read(0,"0\n",1024) = 2
write(1,"lib2->lib1\n",11) = 11
munmap(0x7b2d366f4000,16416) = 0

write(1,"\320\227\320\260\320\263\321\200\321\203\320\266\320\260\320\265\320\274
\320\261\320\270\320\261\320\273\320\270\320\276\321"... ,56) = 56
openat(AT_FDCWD,"/etc/ld.so.cache",O_RDONLY|O_CLOEXEC) = 3
fstat(3,{st_mode=S_IFREG|0644,st_size=60107,...}) = 0
mmap(NULL,60107,PROT_READ,MAP_PRIVATE,3,0) = 0x7b2d366ea000
close(3) = 0

openat(AT_FDCWD,"/lib/x86_64-linux-gnu/libgcd_sin1.so",O_RDONLY|O_CLOEXEC)
= -1 ENOENT (No such file or directory)

openat(AT_FDCWD,"/usr/lib/x86_64-linux-gnu/libgcd_sin1.so",O_RDONLY|O_CLOEXEC)
= -1 ENOENT (No such file or directory)
openat(AT_FDCWD,"/lib/libgcd_sin1.so",O_RDONLY|O_CLOEXEC) = -1
ENOENT (No such file or directory)
openat(AT_FDCWD,"/usr/lib/libgcd_sin1.so",O_RDONLY|O_CLOEXEC) = -1
ENOENT (No such file or directory)
munmap(0x7b2d366ea000,60107) = 0
openat(AT_FDCWD,"./libgcd_sin1.so",O_RDONLY|O_CLOEXEC) = 3

read(3,"\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"... ,832)
= 832
fstat(3,{st_mode=S_IFREG|0775,st_size=15536,...}) = 0
getcwd("/home/alaineredd/workspace/OSLabs/4/build/bin",128) = 46
mmap(NULL,16416,PROT_READ,MAP_PRIVATE|MAP_DENYWRITE,3,0) =
0x7b2d366f4000

mmap(0x7b2d366f5000,4096,PROT_READ|PROT_EXEC,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0)
= 0x7b2d366f5000

mmap(0x7b2d366f6000,4096,PROT_READ,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x2000)
= 0x7b2d366f6000

mmap(0x7b2d366f7000,8192,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0)
= 0x7b2d366f7000
close(3) = 0
mprotect(0x7b2d366f7000,4096,PROT_READ) = 0

```

```

write(1,"\320\221\320\270\320\261\320\273\320\270\320\276\321\202\320\265\320\272\320\
\321\203\321\201\320\277\320\265\321\210\320"... ,73) = 73
    write(1,">",2) = 2
    read(0,"0\n",1024) = 2
    write(1,"lib1->lib2\n",11) = 11
    munmap(0x7b2d366f4000,16416) = 0

write(1,"\320\227\320\260\320\263\321\200\321\203\320\266\320\260\320\265\320\274
\320\261\320\270\320\261\320\273\320\270\320\276\321"... ,56) = 56
    openat(AT_FDCWD,"/etc/ld.so.cache",O_RDONLY|O_CLOEXEC) = 3
    fstat(3,{st_mode=S_IFREG|0644,st_size=60107,...}) = 0
    mmap(NULL,60107,PROT_READ,MAP_PRIVATE,3,0) = 0x7b2d366ea000
    close(3) = 0

openat(AT_FDCWD,"/lib/x86_64-linux-gnu/libgcdc_sin2.so",O_RDONLY|O_CLOEXEC)
= -1 ENOENT (No such file or directory)

openat(AT_FDCWD,"/usr/lib/x86_64-linux-gnu/libgcdc_sin2.so",O_RDONLY|O_CLOEXEC)
= -1 ENOENT (No such file or directory)
    openat(AT_FDCWD,"/lib/libgcdc_sin2.so",O_RDONLY|O_CLOEXEC) = -1
ENOENT (No such file or directory)
    openat(AT_FDCWD,"/usr/lib/libgcdc_sin2.so",O_RDONLY|O_CLOEXEC) = -1
ENOENT (No such file or directory)
    munmap(0x7b2d366ea000,60107) = 0
    openat(AT_FDCWD,"./libgcdc_sin2.so",O_RDONLY|O_CLOEXEC) = 3

read(3,"\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"... ,832)
= 832
    fstat(3,{st_mode=S_IFREG|0775,st_size=15608,...}) = 0
    getcwd("/home/alaineredd/workspace/OSLabs/4/build/bin",128) = 46
    mmap(NULL,16416,PROT_READ,MAP_PRIVATE|MAP_DENYWRITE,3,0) =
0x7b2d366f4000

mmap(0x7b2d366f5000,4096,PROT_READ|PROT_EXEC,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x
= 0x7b2d366f5000

mmap(0x7b2d366f6000,4096,PROT_READ,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x2000)
= 0x7b2d366f6000

mmap(0x7b2d366f7000,8192,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0
= 0x7b2d366f7000
    close(3) = 0
    mprotect(0x7b2d366f7000,4096,PROT_READ) = 0

write(1,"\320\221\320\270\320\261\320\273\320\270\320\276\321\202\320\265\320\272\320\
\321\203\321\201\320\277\320\265\321\210\320"... ,73) = 73
    write(1,">",2) = 2

```

```

    read(0,"1 0 3.14159 0.001\n",1024)      = 18
    write(1,"\342\210\253sin(x)dx \320\276\321\202 0 \320\264\320\276
3.14159 "...,62) = 62
    write(1,"\320\234\320\265\321\202\320\276\320\264:
\321\202\321\200\320\260\320\277\320\265\321\206\320\270\320\270\n",29) =
29
    write(1,">",2)                          = 2
    read(0,"exit\n",1024)                    = 5
    write(1,"\320\262\321\213\321\205\320\276\320\264\n",11) = 11
    munmap(0x7b2d366f4000,16416)             = 0
    exit_group(0)                             = ?
+++ exited with 0 +++

```

Листинг 4: *Strace логи 4.1*

```

execve("./program1",["./program1"],0x7ffd3ce3b010 /* 49 vars */) =
0
    brk(NULL)                                = 0x5cd5ce275000
    mmap(NULL,8192,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_ANONYMOUS,-1,0)
= 0x7cc6a28bb000
    access("/etc/ld.so.preload",R_OK)         = -1 ENOENT (No such file or
directory)

    openat(AT_FDCWD,"/home/alaineredd/workspace/OSLabs/4/build/lib/glibc-hwcaps/x86-64-v3
= -1 ENOENT (No such file or directory)

    newfstatat(AT_FDCWD,"/home/alaineredd/workspace/OSLabs/4/build/lib/glibc-hwcaps/x86-6
= -1 ENOENT (No such file or directory)

    openat(AT_FDCWD,"/home/alaineredd/workspace/OSLabs/4/build/lib/glibc-hwcaps/x86-64-v2
= -1 ENOENT (No such file or directory)

    newfstatat(AT_FDCWD,"/home/alaineredd/workspace/OSLabs/4/build/lib/glibc-hwcaps/x86-6
= -1 ENOENT (No such file or directory)

    openat(AT_FDCWD,"/home/alaineredd/workspace/OSLabs/4/build/lib/libgcd_sin1.so",O_RDONLY
= 3

    read(3,"\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"... ,832)
= 832
    fstat(3,{st_mode=S_IFREG|0775,st_size=15536,...}) = 0
    mmap(NULL,16416,PROT_READ,MAP_PRIVATE|MAP_DENYWRITE,3,0) =
0x7cc6a28b6000

    mmap(0x7cc6a28b7000,4096,PROT_READ|PROT_EXEC,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x
= 0x7cc6a28b7000

```

```

mmap(0x7cc6a28b8000,4096,PROT_READ,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x2000)
= 0x7cc6a28b8000

mmap(0x7cc6a28b9000,8192,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0
= 0x7cc6a28b9000
    close(3)                                = 0

openat(AT_FDCWD,"/home/alaineredd/workspace/OSLabs/4/build/lib/libstdc++.so.6",O_RDONLY
= -1 ENOENT (No such file or directory)
    openat(AT_FDCWD,"/etc/ld.so.cache",O_RDONLY|O_CLOEXEC) = 3
    fstat(3,{st_mode=S_IFREG|0644,st_size=60107,...}) = 0
    mmap(NULL,60107,PROT_READ,MAP_PRIVATE,3,0) = 0x7cc6a28a7000
    close(3)                                = 0

openat(AT_FDCWD,"/lib/x86_64-linux-gnu/libstdc++.so.6",O_RDONLY|O_CLOEXEC)
= 3

read(3,"\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"... ,832)
= 832
    fstat(3,{st_mode=S_IFREG|0644,st_size=2592224,...}) = 0
    mmap(NULL,2609472,PROT_READ,MAP_PRIVATE|MAP_DENYWRITE,3,0) =
0x7cc6a2600000

mmap(0x7cc6a269d000,1343488,PROT_READ|PROT_EXEC,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3
= 0x7cc6a269d000

mmap(0x7cc6a27e5000,552960,PROT_READ,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x1e5000)
= 0x7cc6a27e5000

mmap(0x7cc6a286c000,57344,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,
= 0x7cc6a286c000

mmap(0x7cc6a287a000,12608,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS,-1
= 0x7cc6a287a000
    close(3)                                = 0

openat(AT_FDCWD,"/home/alaineredd/workspace/OSLabs/4/build/lib/libgcc_s.so.1",O_RDONLY
= -1 ENOENT (No such file or directory)

openat(AT_FDCWD,"/lib/x86_64-linux-gnu/libgcc_s.so.1",O_RDONLY|O_CLOEXEC)
= 3

read(3,"\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"... ,832)
= 832
    fstat(3,{st_mode=S_IFREG|0644,st_size=183024,...}) = 0
    mmap(NULL,185256,PROT_READ,MAP_PRIVATE|MAP_DENYWRITE,3,0) =
0x7cc6a25d2000

```

```

mmap(0x7cc6a25d6000,147456,PROT_READ|PROT_EXEC,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0)
= 0x7cc6a25d6000

mmap(0x7cc6a25fa000,16384,PROT_READ,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x28000)
= 0x7cc6a25fa000

mmap(0x7cc6a25fe000,8192,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0)
= 0x7cc6a25fe000
    close(3)                                = 0

openat(AT_FDCWD,"/home/alaineredd/workspace/OSLabs/4/build/lib/libc.so.6",O_RDONLY|O_CLOEXEC)
= -1 ENOENT (No such file or directory)

openat(AT_FDCWD,"/lib/x86_64-linux-gnu/libc.so.6",O_RDONLY|O_CLOEXEC) = 3

read(3,"\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\220\243\2\0\0\0\0\0"...,832)
= 832

pread64(3,"\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"...,784,64)
= 784
    fstat(3,{st_mode=S_IFREG|0755,st_size=2125328,...}) = 0

pread64(3,"\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"...,784,64)
= 784
    mmap(NULL,2170256,PROT_READ,MAP_PRIVATE|MAP_DENYWRITE,3,0) =
0x7cc6a2200000

mmap(0x7cc6a2228000,1605632,PROT_READ|PROT_EXEC,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0)
= 0x7cc6a2228000

mmap(0x7cc6a23b0000,323584,PROT_READ,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x1b0000)
= 0x7cc6a23b0000

mmap(0x7cc6a23ff000,24576,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0)
= 0x7cc6a23ff000

mmap(0x7cc6a2405000,52624,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS,-1,0)
= 0x7cc6a2405000
    close(3)                                = 0

openat(AT_FDCWD,"/lib/x86_64-linux-gnu/libm.so.6",O_RDONLY|O_CLOEXEC) = 3

read(3,"\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"...,832)
= 832
    fstat(3,{st_mode=S_IFREG|0644,st_size=952616,...}) = 0
    mmap(NULL,950296,PROT_READ,MAP_PRIVATE|MAP_DENYWRITE,3,0) =
0x7cc6a24e9000

```

```

mmap(0x7cc6a24f9000,520192,PROT_READ|PROT_EXEC,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0)
= 0x7cc6a24f9000

mmap(0x7cc6a2578000,360448,PROT_READ,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0x8f000)
= 0x7cc6a2578000

mmap(0x7cc6a25d0000,8192,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,3,0)
= 0x7cc6a25d0000
    close(3) = 0
    mmap(NULL,8192,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_ANONYMOUS,-1,0)
= 0x7cc6a28a5000

mmap(NULL,12288,PROT_READ|PROT_WRITE,MAP_PRIVATE|MAP_ANONYMOUS,-1,0) =
0x7cc6a28a2000
    arch_prctl(ARCH_SET_FS,0x7cc6a28a2740) = 0
    set_tid_address(0x7cc6a28a2a10) = 15872
    set_robust_list(0x7cc6a28a2a20,24) = 0
    rseq(0x7cc6a28a3060,0x20,0,0x53053053) = 0
    mprotect(0x7cc6a23ff000,16384,PROT_READ) = 0
    mprotect(0x7cc6a25d0000,4096,PROT_READ) = 0
    mprotect(0x7cc6a25fe000,4096,PROT_READ) = 0
    mprotect(0x7cc6a286c000,45056,PROT_READ) = 0
    mprotect(0x7cc6a28b9000,4096,PROT_READ) = 0
    mprotect(0x5cd5bc580000,4096,PROT_READ) = 0
    mprotect(0x7cc6a28f9000,8192,PROT_READ) = 0

prlimit64(0,RLIMIT_STACK,NULL,{rlim_cur=8192*1024,rlim_max=RLIM64_INFINITY})
= 0
    munmap(0x7cc6a28a7000,60107) = 0
    futex(0x7cc6a287a7bc,FUTEX_WAKE_PRIVATE,2147483647) = 0
    getrandom("\x39\xc4\xfb\x6b\x77\x43\xc3\x21",8,GRND_NONBLOCK) = 8
    brk(NULL) = 0x5cd5ce275000
    brk(0x5cd5ce296000) = 0x5cd5ce296000
    fstat(1,{st_mode=S_IFCHR|0620,st_rdev=makedev(0x88,0x1),...}) = 0
    write(1,">",2) = 2
    fstat(0,{st_mode=S_IFCHR|0620,st_rdev=makedev(0x88,0x1),...}) = 0
    read(0,"1 0 3.14159 0.001\n",1024) = 18
    write(1,"\342\210\253sin(x)dx \320\276\321\202 0 \320\264\320\276
3.14159 "...,56) = 56
    write(1,">",2) = 2
    read(0,"0\n",1024) = 2
    write(1,">",2) = 2
    read(0,"2 6 12\n",1024) = 7
    write(1,"\320\235\320\236\320\224(6,12) = 6\n",18) = 18
    write(1,">",2) = 2
    read(0,"exit\n",1024) = 5
    write(1,"\320\222\321\213\321\205\320\276\320\264 \320\270\320\267
\320\277\321\200\320\276\320\263\321\200\320\260\320\274\320\274"... ,35)

```

```
= 35
exit_group(0)                                = ?
+++ exited with 0 +++
```

Листинг 5: *Strace логи 4.2*